

Air Conditioners Data Analysis

Stage 3: Data Extraction

This notebook is independently executable in Google Colab. Every step explains what we do, why we do it, what we obtain, and how the result is used in the next stage.

Stage Objective

Stage 3 extracts the fields required for analysis and creates focused representations of the product data.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

# Load the project CSV directly so this notebook is independently executable.
csv_path1 = "/content/Most_Expensive_Products.csv"
df1 = pd.read_csv(csv_path1)
csv_path2 = "/content/Top_Discounted_Products.csv"
df2 = pd.read_csv(csv_path2)
csv_path3 = "/content/Top_Rated_Products.csv"
df3 = pd.read_csv(csv_path3)

print("CSV loaded successfully.")
print("Dataset shape:", df1.shape)
print("Dataset shape:", df2.shape)
print("Dataset shape:", df3.shape)
```

```
CSV loaded successfully.
Dataset shape: (15, 14)
Dataset shape: (15, 14)
Dataset shape: (15, 14)
```

Why Extraction Matters

Working with every field at every step makes analysis harder to explain. Selecting relevant fields creates smaller, purpose-specific views.

```
nums=df1.select_dtypes(include=np.number).columns.tolist(); print(nums)
nums=df2.select_dtypes(include=np.number).columns.tolist(); print(nums)
nums=df3.select_dtypes(include=np.number).columns.tolist(); print(nums)
```

```
['ratings', 'no_of_ratings', 'discount_price', 'actual_price', 'discount_amount', 'discount_percent', 'tonnage', 'energy_star']
['ratings', 'no_of_ratings', 'discount_price', 'actual_price', 'discount_amount', 'discount_percent', 'tonnage', 'energy_star']
['ratings', 'no_of_ratings', 'discount_price', 'actual_price', 'discount_amount', 'discount_percent', 'tonnage', 'energy_star']
```

Reload the CSV

The CSV is read directly so this stage remains self-contained.

```
cats=df1.select_dtypes(include=['object','category','bool']).columns.tolist(); print(cats)
cats=df2.select_dtypes(include=['object','category','bool']).columns.tolist(); print(cats)
cats=df3.select_dtypes(include=['object','category','bool']).columns.tolist(); print(cats)
```

```
['name', 'main_category', 'sub_category', 'image', 'link', 'brand']
['name', 'main_category', 'sub_category', 'image', 'link', 'brand']
['name', 'main_category', 'sub_category', 'image', 'link', 'brand']
```

Identify Numeric Fields

The project contains numeric fields such as: **ratings**, **no_of_ratings**, **discount_price**, **actual_price**, **discount_amount**, **discount_percent**, **tonnage**, **energy_star**. These fields can support descriptive statistics and numerical comparisons.

```
numeric_view=df1[nums].copy(); print(numeric_view.shape)
numeric_view=df2[nums].copy(); print(numeric_view.shape)
numeric_view=df3[nums].copy(); print(numeric_view.shape)
```

```
(15, 8)
(15, 8)
(15, 8)
```

Identify Categorical Fields

The project contains categorical/text fields such as: **name**, **main_category**, **sub_category**, **image**, **link**, **brand**. These fields can support grouping and category-level comparisons.

```
categorical_view=df1[cats].copy(); print(categorical_view.shape)
categorical_view=df2[cats].copy(); print(categorical_view.shape)
categorical_view=df3[cats].copy(); print(categorical_view.shape)
```

```
(15, 6)
(15, 6)
(15, 6)
```

Extract Numeric View

A numeric-only view is created to support later statistical analysis without unnecessary text columns.

```
print(numeric_view.head())
print(numeric_view.head())
print(numeric_view.head())
```

```

ratings  no_of_ratings  discount_price  actual_price  discount_amount  \
0      5.0             5.0         58890.0         84990.0         26100.0
1      5.0             5.0         35490.0         54990.0         19500.0
2      5.0             4.0         62250.0         83900.0         21650.0
3      5.0             2.0         43990.0         61400.0         17410.0
4      5.0             2.0          599.0           999.0           400.0

```

```

discount_percent  tonnage  energy_star
0      30.709495      2.0         5.0
1      35.460993      1.0         3.0
2      25.804529      1.8         5.0
3      28.355049      1.5         3.0
4      40.040040      NaN         NaN

```

```

ratings  no_of_ratings  discount_price  actual_price  discount_amount  \
0      5.0             5.0         58890.0         84990.0         26100.0
1      5.0             5.0         35490.0         54990.0         19500.0
2      5.0             4.0         62250.0         83900.0         21650.0
3      5.0             2.0         43990.0         61400.0         17410.0
4      5.0             2.0          599.0           999.0           400.0

```

```

discount_percent  tonnage  energy_star
0      30.709495      2.0         5.0
1      35.460993      1.0         3.0
2      25.804529      1.8         5.0
3      28.355049      1.5         3.0
4      40.040040      NaN         NaN

```

```

ratings  no_of_ratings  discount_price  actual_price  discount_amount  \
0      5.0             5.0         58890.0         84990.0         26100.0
1      5.0             5.0         35490.0         54990.0         19500.0
2      5.0             4.0         62250.0         83900.0         21650.0
3      5.0             2.0         43990.0         61400.0         17410.0
4      5.0             2.0          599.0           999.0           400.0

```

```

discount_percent  tonnage  energy_star
0      30.709495      2.0         5.0
1      35.460993      1.0         3.0
2      25.804529      1.8         5.0
3      28.355049      1.5         3.0
4      40.040040      NaN         NaN

```

Extract Categorical View

A categorical view is created to support frequency counts and group comparisons.

```
print(categorical_view.head())
print(categorical_view.head())
print(categorical_view.head())
```

```

name main_category  \
0  Voltas 2 Ton 5 Star Inverter Split AC (Copper ...  appliances
1  LG 1 Ton 3 Star Split Dual Inverter AC - White...  appliances
2  Daikin 1.8 Ton 5 Star Inverter Split AC (Coppe...  appliances
3  Panasonic 1.5 Ton 3 Star Hot and Cold Wi-Fi In...  appliances
4  AC Stand/Heavy Duty Air Conditioner Outdoor Un...  appliances

```

```

sub_category  image  \
0  Air Conditioners  https://m.media-amazon.com/images/I/31kzjzWD29...
1  Air Conditioners  https://m.media-amazon.com/images/W/IMAGERENDE...
2  Air Conditioners  https://m.media-amazon.com/images/I/41zljgtpK...
3  Air Conditioners  https://m.media-amazon.com/images/I/71z2BRd3jb...
4  Air Conditioners  https://m.media-amazon.com/images/W/IMAGERENDE...

```

```

link  brand
0  https://www.amazon.in/Voltas-Inverter-Condense...  Voltas
1  https://www.amazon.in/LG-Star-Split-Dual-Inver...  Lg
2  https://www.amazon.in/Daikin-Inverter-Split-Co...  Daikin
3  https://www.amazon.in/Panasonic-Convertible-ad...  Panasonic
4  https://www.amazon.in/Stand-Conditioner-Outdoo...  Ac

```

```

name main_category  \
0  Voltas 2 Ton 5 Star Inverter Split AC (Copper ...  appliances
1  LG 1 Ton 3 Star Split Dual Inverter AC - White...  appliances
2  Daikin 1.8 Ton 5 Star Inverter Split AC (Coppe...  appliances
3  Panasonic 1.5 Ton 3 Star Hot and Cold Wi-Fi In...  appliances
4  AC Stand/Heavy Duty Air Conditioner Outdoor Un...  appliances

```

```

sub_category  image  \
0  Air Conditioners  https://m.media-amazon.com/images/I/31kzjzWD29...
1  Air Conditioners  https://m.media-amazon.com/images/W/IMAGERENDE...
2  Air Conditioners  https://m.media-amazon.com/images/I/41zljgtpK...
3  Air Conditioners  https://m.media-amazon.com/images/I/71z2BRd3jb...
4  Air Conditioners  https://m.media-amazon.com/images/W/IMAGERENDE...

```

```

link  brand
0  https://www.amazon.in/Voltas-Inverter-Condense...  Voltas
1  https://www.amazon.in/LG-Star-Split-Dual-Inver...  Lg
2  https://www.amazon.in/Daikin-Inverter-Split-Co...  Daikin
3  https://www.amazon.in/Panasonic-Convertible-ad...  Panasonic
4  https://www.amazon.in/Stand-Conditioner-Outdoo...  Ac

```

```

name main_category  \
0  Voltas 2 Ton 5 Star Inverter Split AC (Copper ...  appliances
1  LG 1 Ton 3 Star Split Dual Inverter AC - White...  appliances
2  Daikin 1.8 Ton 5 Star Inverter Split AC (Coppe...  appliances
3  Panasonic 1.5 Ton 3 Star Hot and Cold Wi-Fi In...  appliances
4  AC Stand/Heavy Duty Air Conditioner Outdoor Un...  appliances

```

```

sub_category  image  \
0  Air Conditioners  https://m.media-amazon.com/images/I/31kzjzWD29...
1  Air Conditioners  https://m.media-amazon.com/images/W/IMAGERENDE...
2  Air Conditioners  https://m.media-amazon.com/images/I/41zljgtpK...
3  Air Conditioners  https://m.media-amazon.com/images/I/71z2BRd3jb...
4  Air Conditioners  https://m.media-amazon.com/images/W/IMAGERENDE...

```

```

link  brand
0  https://www.amazon.in/Voltas-Inverter-Condense...  Voltas
1  https://www.amazon.in/LG-Star-Split-Dual-Inver...  Lg
2  https://www.amazon.in/Daikin-Inverter-Split-Co...  Daikin

```

Inspect Extracted Views

Shapes and sample records are displayed so we can confirm that extraction preserved the intended observations.

```
for c in nums[:5]: print(c, numeric_view[c].describe().to_dict())
for c in nums[:5]: print(c, numeric_view[c].describe().to_dict())
for c in nums[:5]: print(c, numeric_view[c].describe().to_dict())

ratings {'count': 15.0, 'mean': 5.0, 'std': 0.0, 'min': 5.0, '25%': 5.0, '50%': 5.0, '75%': 5.0, 'max': 5.0}
no_of_ratings {'count': 15.0, 'mean': 2.4, 'std': 1.2421180068162376, 'min': 1.0, '25%': 2.0, '50%': 2.0, '75%': 2.0, 'max': 5.0}
discount_price {'count': 15.0, 'mean': 48072.6, 'std': 26957.0542577846, 'min': 599.0, '25%': 39190.0, '50%': 39190.0, '75%': 53895.0, 'max': 128800.0}
actual_price {'count': 15.0, 'mean': 66387.26666666666, 'std': 30467.752588285086, 'min': 999.0, '25%': 56990.0, '50%': 56990.0, '75%': 77900.0, 'max': 149000.0}
discount_amount {'count': 15.0, 'mean': 18314.666666666668, 'std': 7008.674081518729, 'min': 400.0, '25%': 17800.0, '50%': 17800.0, '75%': 21280.0, 'max': 29660.0}
ratings {'count': 15.0, 'mean': 5.0, 'std': 0.0, 'min': 5.0, '25%': 5.0, '50%': 5.0, '75%': 5.0, 'max': 5.0}
no_of_ratings {'count': 15.0, 'mean': 2.4, 'std': 1.2421180068162376, 'min': 1.0, '25%': 2.0, '50%': 2.0, '75%': 2.0, 'max': 5.0}
discount_price {'count': 15.0, 'mean': 48072.6, 'std': 26957.0542577846, 'min': 599.0, '25%': 39190.0, '50%': 39190.0, '75%': 53895.0, 'max': 128800.0}
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discount_amount {'count': 15.0, 'mean': 18314.666666666668, 'std': 7008.674081518729, 'min': 400.0, '25%': 17800.0, '50%': 17800.0, '75%': 21280.0, 'max': 29660.0}
ratings {'count': 15.0, 'mean': 5.0, 'std': 0.0, 'min': 5.0, '25%': 5.0, '50%': 5.0, '75%': 5.0, 'max': 5.0}
no_of_ratings {'count': 15.0, 'mean': 2.4, 'std': 1.2421180068162376, 'min': 1.0, '25%': 2.0, '50%': 2.0, '75%': 2.0, 'max': 5.0}
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actual_price {'count': 15.0, 'mean': 66387.26666666666, 'std': 30467.752588285086, 'min': 999.0, '25%': 56990.0, '50%': 56990.0, '75%': 77900.0, 'max': 149000.0}
discount_amount {'count': 15.0, 'mean': 18314.666666666668, 'std': 7008.674081518729, 'min': 400.0, '25%': 17800.0, '50%': 17800.0, '75%': 21280.0, 'max': 29660.0}
```

Derived Analytical Views

Where appropriate, focused combinations of fields can be used for specific questions, such as comparing a product's price with its rating or category.

```
for c in cats[:5]: print(c, categorical_view[c].nunique(dropna=False))
for c in cats[:5]: print(c, categorical_view[c].nunique(dropna=False))
for c in cats[:5]: print(c, categorical_view[c].nunique(dropna=False))
```

```
name 15
main_category 1
sub_category 1
image 15
link 15
name 15
main_category 1
sub_category 1
image 15
link 15
name 15
main_category 1
sub_category 1
image 15
link 15
```

What We Obtained

The output of this stage is a set of clearly defined analytical views rather than an unstructured collection of columns.

```
print('Extraction stage completed.')

Extraction stage completed.
```